

BANARAS HINDU UNIVERSITY

B.Sc. (Hons) Semester III Examination, 2025-26 (NEP)

PHYSICS**Paper Code: PHYMV31 : Applied Electronics and Fiber Optics**

Ref. No.: CECONF/N/2025-26/009

Time: $2\frac{1}{2}$ Hours

Maximum Marks: 70

Instructions: Answer five questions including Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer all questions briefly: $[10 \times 2 = 20]$

- (a) What is multimeter? What are its uses?
- (b) What is Digital Storage Oscilloscope (DSO)? State one advantage over Cathode Ray Oscilloscope (CRO).
- (c) Differentiate between active and passive electronic components.
- (d) Describe briefly the depletion layer in a p-n junction diode.
- (e) What is field-effect transistor? Mention one of its advantages over a bipolar junction transistor.
- (f) A transistor has a current gain $\beta = 100$. Find the value of current gain α .
- (g) A transformer has 200 turns on its primary coil and 800 turns on its secondary coil. If the primary voltage is 50 V, calculate the secondary voltage of the transformer.
- (h) Write two applications of optical fibers.
- (i) An optical fiber has a core refractive index $n_1 = 1.500$ and a cladding refractive index $n_2 = 1.414$. Find numerical aperture and acceptance angle in air.
- (j) What is a Laser and why is it used in optical fiber communication?

Question 2.

- (a) What is a galvanometer? Explain the conversion of a galvanometer into a voltmeter and an ammeter. $[6]$
- (b) A galvanometer has an internal resistance of 50Ω and the full-scale deflection current is 2 mA. We want to convert it into a 5-volt voltmeter. Find the value of the series resistance. $[5+3]$
- (c) What is the difference between a step-up and a step-down transformer? $[4\frac{1}{2}]$

Question 3.

- (a) Explain the construction, working, and applications of a cathode ray oscilloscope (CRO). $[6]$
- (b) With the help of a neat circuit diagram, discuss the working of a Zener diode as a voltage regulator. $[6\frac{1}{2}]$

Question 4.

- (a) With the help of a neat circuit diagram, explain the working of a full-wave rectifier. Find the efficiency and the ripple factor of a full-wave rectifier. [7]
- (b) Write the current-voltage ($I - V$) equation of a p-n junction diode and obtain the following expression for the dynamic forward resistance R_f of the diode: $R_f = \frac{\eta kT}{eI}$. [5½]

Question 5.

- (a) What is a light-emitting diode? Explain briefly its construction and working. [6]
- (b) Draw a neat circuit diagram showing an n-p-n transistor as an amplifier in the common emitter (CE) mode. What is the meaning of the phase shift in the CE mode? [6½]

Question 6.

- (a) Explain the mechanism of total internal reflection (TIR) in optical fibers. Derive the expression for numerical aperture (NA) and discuss how NA influences light-gathering ability. [8]
- (b) Explain the terms emitter, base, and collector in the transistors. [4½]

Question 7.

- (a) Describe the different types of optical fiber losses, including absorption, scattering, bending loss and modal dispersion. Explain the physical origin of each type and discuss methods to minimize these losses in practical fiber communication systems. [7]
- (b) Explain the difference between the Avalanche breakdown and the Zener breakdown. [5½]
